

# Modelleme ve Benzetim

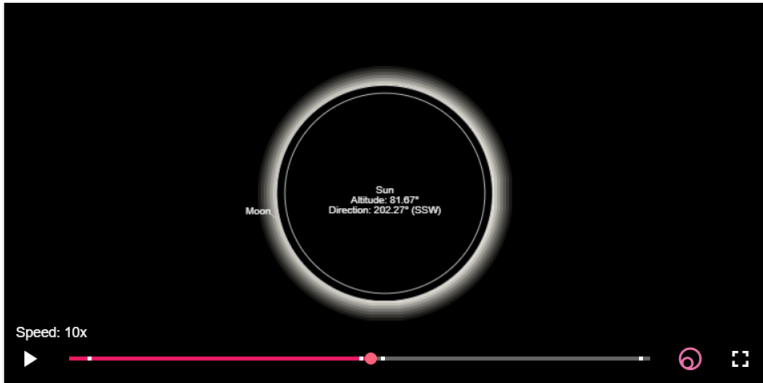
Rahmi Elibol

24.09.2026

Osmaniye

# Tam Güneş tutulması

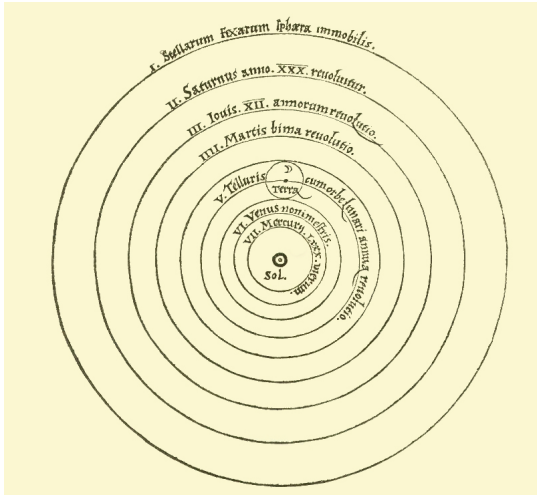
2 Ağustos 2027



1

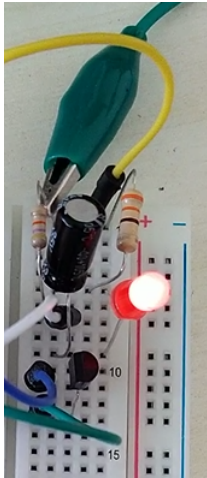
<sup>1</sup><https://www.timeanddate.com/eclipse/solar/2027-august-2>

# Model

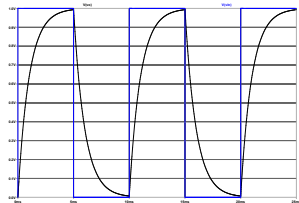
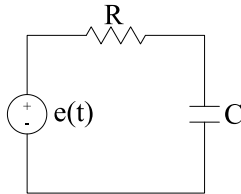


<sup>2</sup>Kopernik, N., 'De revolutionibus orbium coelestium', 1543

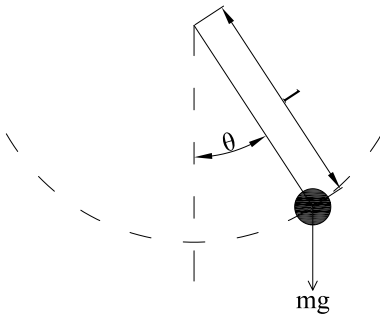
## RC Devresi



$$\frac{dv_C}{dt} = \frac{-v_C + e(t)}{RC}$$



## Basit sarkaç



Sarkaç denklemi:

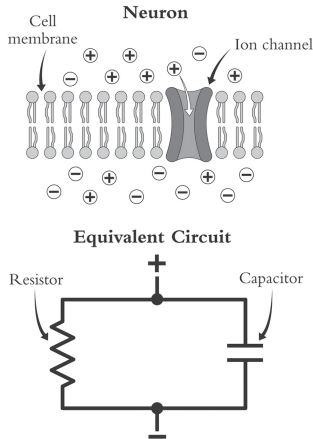
$$ml\ddot{\theta} = -mg\sin\theta - k l \theta$$

$x_1 = \theta$  ve  $x_2 = \dot{\theta}$  olarak düzenlersek;

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\frac{g}{l} \sin x_1 - \frac{k}{m} x_2$$

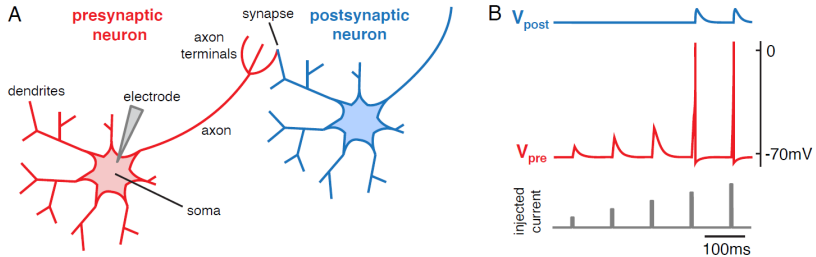
## Hücre zarı ve eşdeğer devresi



3

<sup>3</sup>Lindsay G., Models of the Mind: How Physics, Engineering and Mathematics Have Shaped Our Understanding of the Brain, Bloomsbury Publishing, 2021.

## Sinir hücreleri ve bağlantıları



4

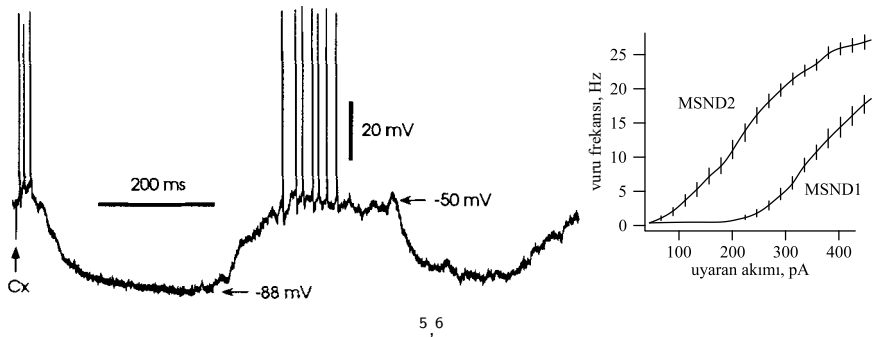
Model: Topla ve ateşle modeli (Integrate and Fire)

$$\tau_m \frac{dv_m}{dt} = -(v_m - E_L) + I_{ic}$$

$$v_m(t) > V_{th} \quad (\text{t anında eylem potansiyeli üretildiğinde}) \quad v_m(t) \leftarrow V_{reset}$$

<sup>4</sup>Robert Rosenbaum, Modeling Neural Circuits Made Simple, 2022

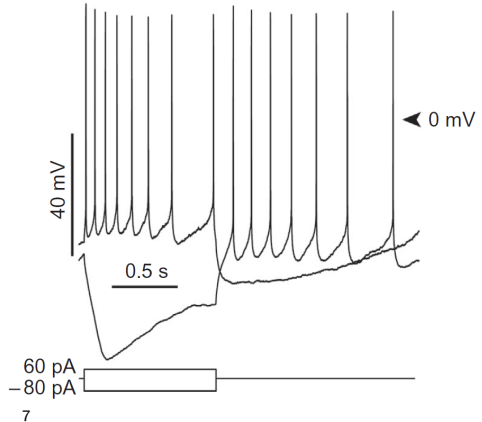
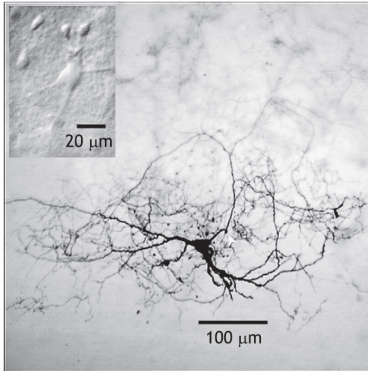
## Ortaboy dikensi sinir hücresi, MSN



<sup>5</sup>Wilson, C.J., Chapter 18 The generation of natural firing patternsin neostriatal neurons, G. Arbuthnott ve P. Emson, (düzenleyenler), Chemical Signalling in the Basal Ganglia, cilt 99 of Progress in Brain Research, Elsevier, s.277 - 297, 1993.

<sup>6</sup>Gertler TS, Chan CS, Surmeier DJ. Dichotomous anatomical properties of adult striatal medium spiny neurons. J Neurosci. 2008 Oct 22;28(43):10814-24. doi: 10.1523/JNEUROSCI.2660-08.2008. PMID: 18945889; PMCID: PMC3235748.

## Ara hücreler, IN



<sup>7</sup>J.A. Goldberg, C.J. Wilson, 2016, Chapter 7 - The Cholinergic Interneuron of the Striatum, Editor(s): Heinz Steiner, Kuei Y. Tseng, Handbook of Behavioral Neuroscience, Elsevier, Volume 24, Pages 137-155, ISSN 1569-7339.



## Izhikevich Modeli, 2003

$$\begin{aligned}\dot{v} &= 0.04v^2 + 5v + 140 - u + I_s \\ \dot{u} &= a(bv - u)\end{aligned}\tag{1}$$

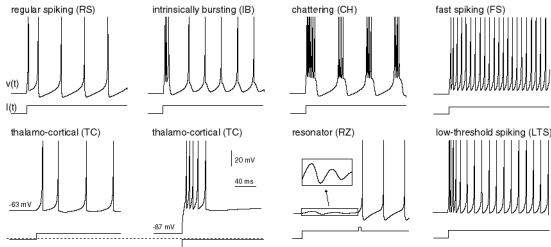
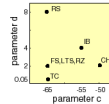
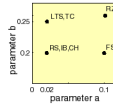
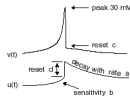
$$v > 30 \text{ mV}, \quad \text{ise} \quad \begin{cases} v \leftarrow c \\ u \leftarrow u + d \end{cases}\tag{2}$$

# Izhikevich Modeli, 2003

$$v' = 0.04v^2 + 5v + 140 - u + I$$

$$u' = a(bv - u)$$

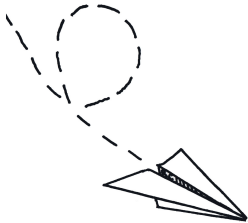
if  $v = 30$  mV,  
then  $v \leftarrow c$ ,  $u \leftarrow u + d$



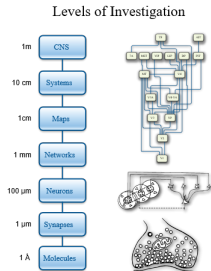
10

<sup>10</sup> Izhikevich, E., 'Which Model to Use for Cortical Spiking Neurons?', IEEE Transactions on Neural Networks (2004) 15:1063-1070

## Modelleme yaklaşımları



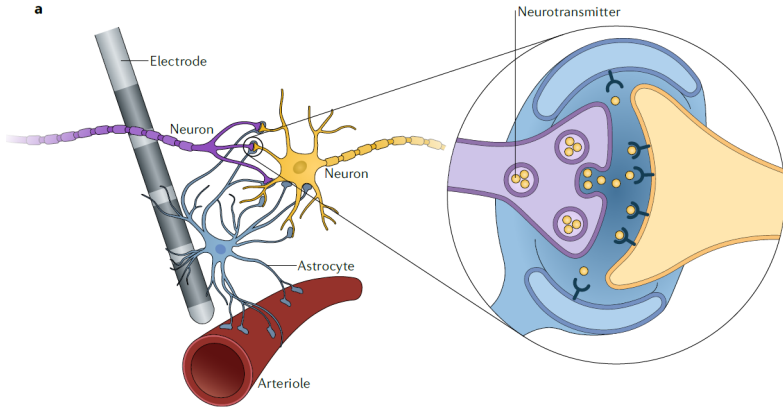
"All models are wrong, but some models are useful." – George Box



<https://cnl.salk.edu/>



## Sinapslar ve sinirileticileri



11

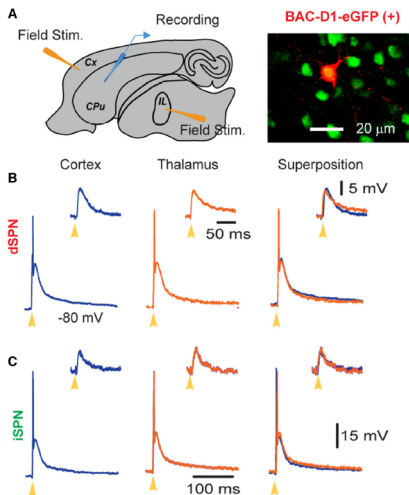
<sup>11</sup>Lozano, A.M., Lipsman, N., Bergman, H. et al. Deep brain stimulation: current challenges and future directions. *Nat Rev Neurol* 15, 148–160 (2019). <https://doi.org/10.1038/s41582-018-0128-2>

# Sinaps modeli

DOI 10.1007/s00429-017-1536-6

ORIGINAL ARTICLE

## Functional comparison of corticostriatal and thalamo postsynaptic responses in striatal neurons of the mouse

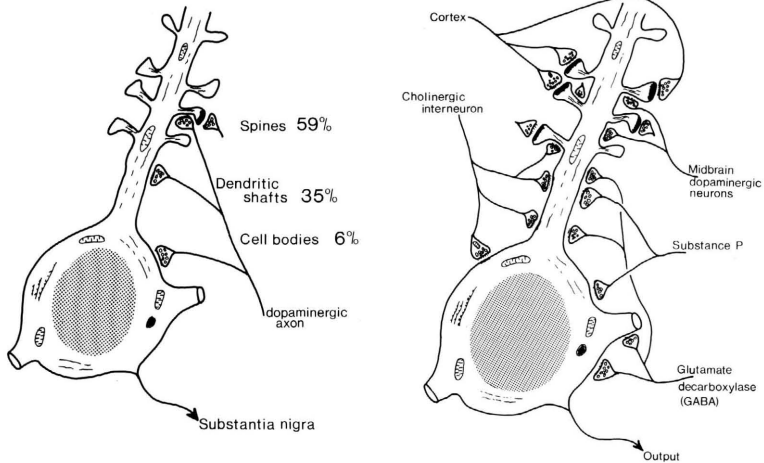
M. A. Arias-García<sup>1</sup> · D. Tapia<sup>1</sup> · J. A. Laville<sup>1</sup> · V. M. Calderón<sup>1</sup> ·  
Y. Ramiro-Cortés<sup>1</sup> · J. Bargas<sup>1</sup> · E. Galarraga<sup>1</sup>Received: 22 March 2017 / Accepted: 5 October 2017  
© Springer-Verlag GmbH Germany 2017

12

<sup>12</sup>Arias-García, M.A., Tapia, D., Laville, J.A. et al., 'Functional comparison of corticostriatal and thalamostriatal postsynaptic responses in striatal neurons of the mouse', *Brain Struct Funct* (2017).  
<https://doi.org/10.1007/s00429-017-1536-6>



## Plastisite

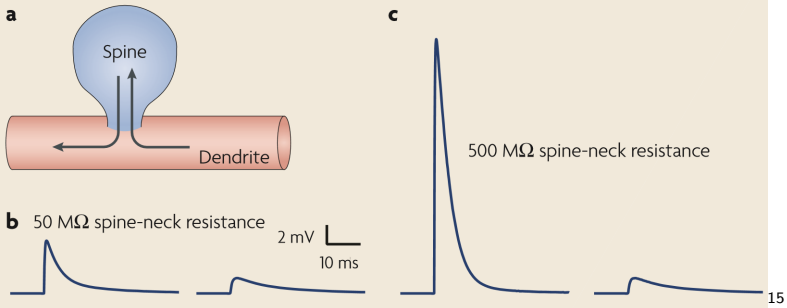


14

<sup>14</sup>Smith AD, Bolam JP, 1990, The neural network of the basal ganglia as revealed by the study of synaptic connections of identified neurones, Trends Neurosci. 1990 Jul;13(7):259-65. doi: 10.1016/0166-2236(90)90106-k

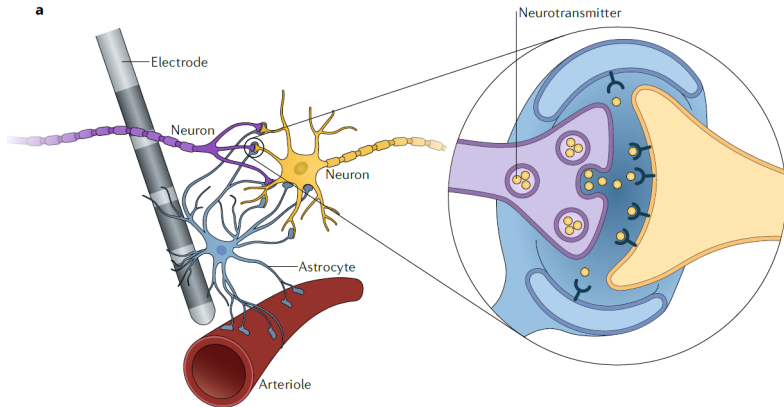
# Plastisite

## Box 1 | Possible functions of dendritic spines



<sup>15</sup>Spruston N., 2008, Pyramidal neurons: dendritic structure and synaptic integration, Nat Rev Neurosci. 2008 Mar;9(3):206-21. doi: 10.1038/nrn2286.

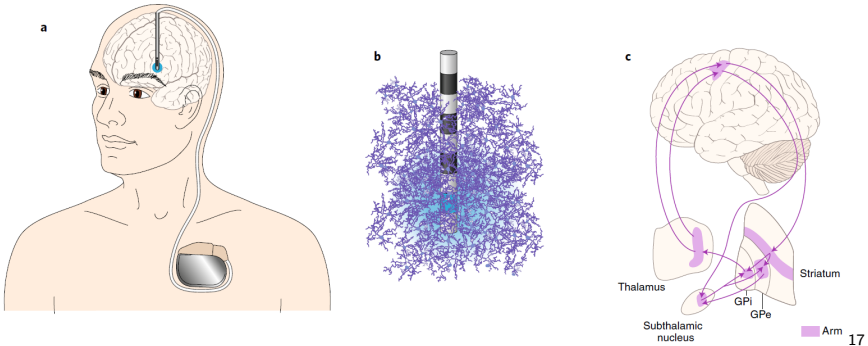
## Plastisite



16

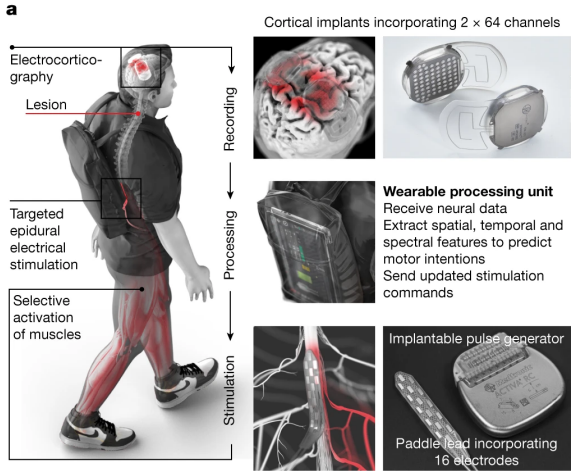
<sup>16</sup>Lozano, A.M., Lipsman, N., Bergman, H. et al. Deep brain stimulation: current challenges and future directions. *Nat Rev Neurol* 15, 148–160 (2019). <https://doi.org/10.1038/s41582-018-0128-2>

## Nerede kullanıyoruz?



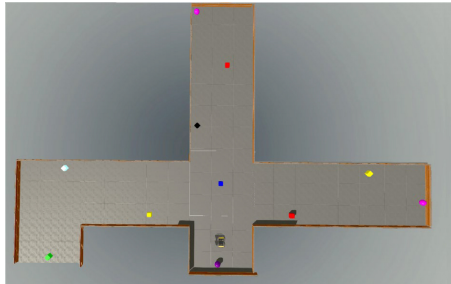
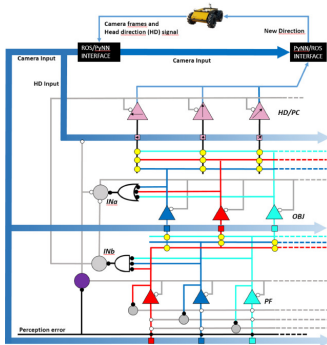
<sup>17</sup>Cagnan, H., Denison, T., McIntyre, C. et al. Emerging technologies for improved deep brain stimulation. Nat Biotechnol 37, 1024–1033 (2019). <https://doi.org/10.1038/s41587-019-0244-6>.

## Nerede kullanıyoruz?



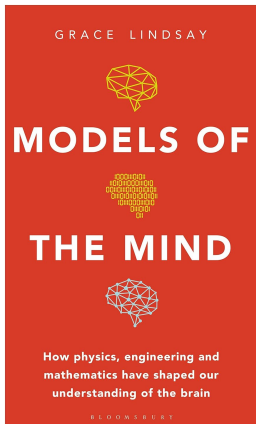
<sup>18</sup>Lorach, H., Galvez, A., Spagnolo, V. et al. Walking naturally after spinal cord injury using a brain–spine interface. Nature 618, 126–133 (2023). <https://doi.org/10.1038/s41586-023-06094-5>.

## Nerede kullanıyoruz?



19

<sup>19</sup> Simone Coppolino, Michele Migliore, An explainable artificial intelligence approach to spatial navigation based on hippocampal circuitry, *Neural Networks*, Volume 163, 2023, Pages 97-107, <https://doi.org/10.1016/j.neunet.2023.03.030>.



Of course, everything in this book may be wrong. It may be wrong because it is science and our understanding of the world is ever-evolving. It may be wrong because it is history and there is always more than one way to tell a story. And, most importantly, it is wrong because it is mathematics. Mathematical models of the mind do not make for perfect replicas of the brain, nor should we strive for them to be. Yet in the study of the most complex object in the known universe, mathematical models are not just useful but absolutely essential. The brain will not be understood through words alone.

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<sup>20</sup>Lindsay G., Models of the Mind: How Physics, Engineering and Mathematics Have Shaped Our Understanding of the Brain, Bloomsbury Publishing, 2021.

Model  
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Hücreler  
○○○○○○○

Sinapslar  
○○○

Plastisite  
○○○○○○

Sonuç  
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Teşekkürler.